

ASSIGNMENT – INDUSTRY-ORIENTED DATA HANDLING AND VISUALIZATION

DSA06 / DATA HANDLING AND VISUALIZATION

PARTICULAR	DETAILS
Department	B. Tech AI&DS
Program	BE
Course Code & Course Name	DSA06 / DATA HANDLING AND VISUALIZATION
Academic Year / Batch	3 rd year (2024 Batch)
Faculty Name	Dr. S. Loganayagi
Assignment Title	Healthcare Service and Patient Data Visualization Analytics System
Date of Issue	
Date of Submission	
Maximum Marks	100

COURSE OUTCOMES

CO1: Explain core concepts of data visualization including data types, aesthetic mappings, scales, coordinate systems, and effective use of color.

CO2: Select and construct appropriate visualizations for representing amounts, distributions, and proportions.

CO3: Visualize and interpret relationships using scatterplots, correlation analysis, and dimensionality reduction techniques such as PCA, t-SNE, and UMAP.

ASSESSMENT PROFILE

ASSESSMENT PROFILE	DETAILS
Bloom's Taxonomy Level	L4 – Analyze / L5 – Evaluate / L6 – Create
SDG Mapping	SDG 3 – Good Health and Well-Being; SDG 9 – Industry, Innovation and Infrastructure

ASSESSMENT PROFILE	DETAILS
Industry / Societal Relevance	Healthcare organizations collect large volumes of patient and service-related data. Appropriate data handling and visualization can help healthcare administrators understand patient characteristics, service utilization, distributions, relationships, and multivariate patterns.

1. Assignment Problem / Challenge

Industry Case Study: Healthcare Service and Patient Data Visualization Analytics System

A healthcare organization manages several hospitals and medical service centers. Every day, large volumes of data are generated from patient visits, diagnostic tests, hospital departments, treatment records, waiting times, service charges, and patient feedback.

The organization stores this information in databases and spreadsheets. However, administrators and healthcare analysts find it difficult to interpret the large amount of available information and identify meaningful patterns using only tabular reports.

The organization requires an effective data visualization system to transform raw healthcare service data into meaningful visual information.

The selected dataset may contain variables such as:

- Patient ID or anonymized record ID.
- Age.
- Gender.
- Department.
- Service type.
- Number of visits.
- Waiting time.
- Treatment cost.
- Number of diagnostic tests.
- Length of hospital stay.
- Patient satisfaction score.
- Medical service category.

This approach creates several challenges.

1. Difficulty Understanding Large Volumes of Healthcare Data

Large healthcare datasets contain numerous observations and variables. Tables alone make it difficult to identify important patterns and variations.

2. Difficulty Selecting Appropriate Visualizations

Healthcare data contains both numerical and categorical variables. Different types of information require different visual representations.

For example:

- Department names are categorical.
- Waiting time is numerical.
- Number of visits is discrete.
- Treatment cost is continuous.

The organization requires appropriate visualization selection based on the characteristics of the data.

3. Limited Understanding of Service Amounts

Administrators need to compare:

- Number of patients visiting each department.
- Number of diagnostic tests performed.
- Number of patients using different services.
- Department-wise service utilization.

These comparisons are difficult to understand from large tables.

4. Difficulty Understanding Patient and Service Distributions

The organization needs to identify:

- Distribution of patient ages.
- Distribution of waiting times.
- Distribution of treatment costs.
- Variation in hospital stay duration.
- Possible outliers.

5. Difficulty Comparing Service Proportions

Healthcare administrators need to understand the contribution of:

- Different departments.

- Different service categories.
- Different patient groups.
- Different treatment types.

6. Difficulty Identifying Relationships

The organization needs to investigate possible relationships between variables such as:

- Waiting time and patient satisfaction.
- Number of diagnostic tests and treatment cost.
- Length of stay and treatment cost.
- Number of visits and healthcare expenditure.

7. Difficulty Understanding Multivariate Patient Data

Healthcare datasets may contain multiple numerical variables. It becomes difficult to identify similarities and patterns among observations when many variables are analyzed simultaneously.

You are appointed as a **Healthcare Data Visualization Analyst** to design and develop a prototype **Healthcare Service and Patient Data Visualization Analytics System**.

The proposed system must transform raw healthcare service data into meaningful visual representations and analytical insights.

CENTRAL HIGHER-ORDER CHALLENGE

Design, implement, experimentally evaluate, and justify a data visualization and analytics system for healthcare service and patient data. The solution must apply appropriate data types, aesthetic mappings, scales, coordinate systems, and color; construct suitable visualizations for amounts, distributions, and proportions; analyze relationships and correlations; and apply dimensionality reduction techniques to multivariate healthcare data. Students must compare alternative visualization and dimensionality reduction approaches and justify their final analytical decisions using evidence obtained from the dataset.

2. Problem Statement

Develop a Python-based **Healthcare Service and Patient Data Visualization Analytics System**.

The system must support the following major components:

MODULE	KEY REQUIREMENTS
Data Handling and Visualization Fundamentals	Load and inspect healthcare data and identify appropriate data types and visual mappings
Amount Visualization	Represent and compare healthcare service quantities
Distribution Analysis	Analyze distributions of numerical healthcare variables
Proportion Visualization	Represent proportions among departments and service categories
Relationship Analysis	Identify associations between healthcare variables
Correlation Analysis	Analyze relationships among multiple numerical variables
Dimensionality Reduction	Apply PCA and compare it with t-SNE or UMAP
Visualization Comparison and Interpretation	Compare visualization approaches and justify analytical choices

The proposed system should allow the analysis of a suitable anonymized healthcare dataset and generate meaningful visual outputs.

3. Requirements and Constraints

3.1 Functional Requirements

The system shall:

- Load a suitable healthcare or patient service dataset.
- Inspect the dataset structure.
- Identify numerical variables.
- Identify categorical variables.
- Identify discrete and continuous variables where appropriate.
- Apply appropriate visual aesthetics.
- Generate amount-based visualizations.
- Generate distribution visualizations.
- Generate proportion visualizations.

- Generate relationship and association visualizations.
 - Perform correlation analysis.
 - Apply PCA for dimensionality reduction.
 - Apply either t-SNE or UMAP as an additional dimensionality reduction technique.
 - Visualize the reduced-dimensional data.
 - Compare visualization techniques.
 - Interpret and document important patterns.
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3.2 Technical Constraints

- Use **Python 3.x** as the implementation language.
 - Use appropriate Python libraries for data handling and visualization.
 - Use a healthcare dataset containing sufficient numerical and categorical variables.
 - The dataset should be anonymized or should not expose unnecessary personally identifiable information.
 - Variables must be appropriately classified before visualization.
 - All major visualizations must contain meaningful titles.
 - Axes must be labeled appropriately.
 - Legends must be included when required.
 - Appropriate visual aesthetics must be selected.
 - PCA is mandatory.
 - Either t-SNE or UMAP must be used for comparison.
 - The dataset source must be documented.
 - The assignment scope must remain strictly within **CO1, CO2, and CO3**.
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3.3 Performance and Reliability Constraints

The system should:

- Process datasets of different sizes where possible.
- Produce reproducible outputs.
- Document important analysis parameters.
- Handle missing or unsuitable values before analysis.
- Generate interpretable outputs.
- Record observations and results for validation.

Students should test the implementation using increasing dataset sizes wherever feasible.

3.4 Data Integrity and Professional Considerations

Students must ensure:

- Healthcare data is handled responsibly.
 - Patient-identifying information is not unnecessarily displayed.
 - The dataset source is documented.
 - Data is not manipulated to force a particular conclusion.
 - Visual and analytical conclusions are supported by available evidence.
 - Observed relationships are not interpreted beyond what the data supports.
-

4. Student Work & Assessment Components

COMPONENT	MARKS	EXPECTED EVIDENCE / OUTPUT
1. Problem Understanding and Formulation	10	Understanding of the healthcare data problem, dataset characteristics, requirements, assumptions, and constraints
2. Application of Course Knowledge	20	Meaningful application of CO1, CO2, and CO3 concepts
3. Solution / Design / Methodology	20	Visualization workflow, data mapping, visualization design, and analytical methodology
4. Use of Modern Tools / Techniques	10	Python environment, visualization libraries, dimensionality reduction tools, and implementation evidence
5. Results, Testing and Validation	15	Test cases, generated visualizations, results, and validation evidence
6. Analysis, Trade-offs and Justification	15	Comparison of techniques and

COMPONENT	MARKS	EXPECTED EVIDENCE / OUTPUT
		evidence-based analytical decisions
7. Broader Considerations / Professional Responsibility	5	Data integrity, privacy, and responsible handling of healthcare data
8. Technical Documentation & Reflection	5	Report quality, references, screenshots, and student reflection
TOTAL	100	

5. Solution / Design / Methodology

5.1 Data Handling and Visualization Fundamentals – CO1

Students must first inspect and understand the selected healthcare dataset.

The following should be identified:

- Numerical variables.
- Categorical variables.
- Discrete variables.
- Continuous variables.

Students must demonstrate appropriate mapping of data to visual aesthetics.

The following aesthetics should be considered where appropriate:

- Position.
- Color.
- Size.
- Shape.

Students must also select suitable:

- Scales.
- Axes.
- Coordinate systems.
- Color representations.

Students should justify why particular visual mappings are suitable for selected healthcare variables.

For example:

- Department → Color.
 - Waiting time → Position.
 - Number of visits → Size.
 - Service category → Shape.
-

5.2 Amount Visualization – CO2

Students must visualize healthcare service amounts using appropriate techniques.

Implement at least **three** of the following:

- Bar plot.
- Grouped bar chart.
- Stacked bar chart.
- Dot plot.
- Heatmap.

Possible analyses include:

- Number of patients by department.
- Number of service visits by category.
- Diagnostic tests performed by department.
- Comparison of patient visits across service categories.

Students must compare at least two suitable visualization techniques for a selected amount-based dataset and justify which technique provides clearer interpretation.

5.3 Distribution Analysis – CO2

Students must analyze the distribution of selected numerical healthcare variables.

Implement at least **four** of the following:

- Histogram.
- Density plot.
- Boxplot.
- Violin plot.
- Empirical Cumulative Distribution Function (ECDF).

- Quantile–Quantile (Q–Q) plot.

Possible variables include:

- Patient age.
- Waiting time.
- Treatment cost.
- Length of stay.
- Patient satisfaction score.

Students should identify:

- Distribution patterns.
- Spread.
- Skewness.
- Outliers.
- Differences between patient or service groups.

Students must compare at least two visualizations representing the same distribution.

5.4 Proportion Visualization – CO2

Students must represent healthcare service proportions.

Implement at least **two** of the following:

- Pie chart.
- Side-by-side bar chart.
- Stacked bar chart.
- Treemap.

Possible analyses include:

- Proportion of patients by department.
- Service category contribution.
- Patient group distribution.
- Treatment category proportions.

Students must compare at least two visualization approaches for the same proportional data.

The final choice must be justified based on the clarity of comparison.

5.5 Relationship and Association Analysis – CO3

Students must investigate relationships between selected healthcare variables.

Implement at least **three** of the following:

- Scatterplot.
- Bubble chart.
- Correlogram.
- Correlation matrix.
- Hexbin plot.

Possible relationships include:

- Waiting time versus patient satisfaction.
- Number of tests versus treatment cost.
- Length of stay versus treatment cost.
- Number of visits versus healthcare expenditure.

Students must interpret:

- Positive associations.
- Negative associations.
- Strong associations.
- Weak associations.

Interpretations must be supported by the visual results.

5.6 Correlation Analysis – CO3

Students must perform correlation analysis using suitable numerical healthcare variables.

The analysis shall include:

- Correlation calculation.
- Correlation matrix.
- Visual representation using a heatmap or correlogram.
- Identification of important relationships.

Students should identify:

- Strong positive correlations.
- Strong negative correlations.
- Weak correlations.

The interpretation should explain the observed numerical and visual relationships.

5.7 Dimensionality Reduction – CO3

Mandatory Technique: Principal Component Analysis (PCA)

Students must:

1. Select appropriate numerical healthcare variables.
2. Prepare the variables appropriately.
3. Apply PCA.
4. Reduce the dataset to a lower-dimensional representation.
5. Visualize the resulting components.

Possible numerical variables may include:

- Age.
- Number of visits.
- Waiting time.
- Number of diagnostic tests.
- Treatment cost.
- Length of stay.
- Satisfaction score.

Comparison Technique

Students must additionally implement either:

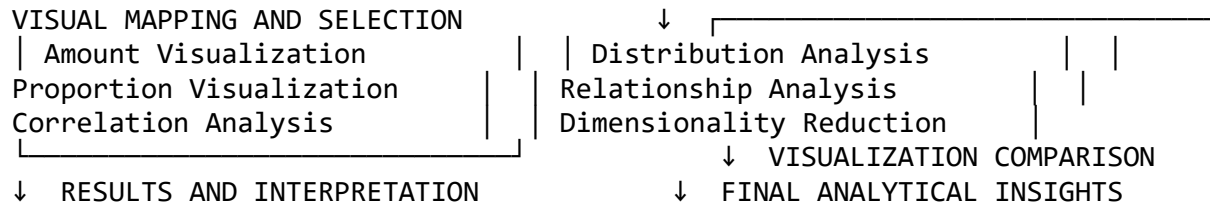
- **t-SNE**, or
- **UMAP**.

Students must compare PCA with the selected technique based on:

- Representation of patterns.
 - Separation of observations.
 - Identification of possible groups.
 - Ease of interpretation.
 - Differences observed in the visual outputs.
-

5.8 Overall Solution Workflow

```
text id="1mpqad" RAW HEALTHCARE DATASET          ↓          DATA LOADING
↓          DATA INSPECTION          ↓ DATA TYPE IDENTIFICATION          ↓
```



6. Results, Testing and Validation

Students must validate the system using suitable test cases.

TEST CASE	SCENARIO	EXPECTED RESULT
TC1	Load a valid healthcare dataset	Dataset loads successfully
TC2	Identify numerical variables	Numerical variables are correctly identified
TC3	Identify categorical variables	Categorical variables are correctly identified
TC4	Generate amount visualization	Healthcare service quantities are clearly represented
TC5	Generate distribution visualization	Distribution characteristics are visible
TC6	Generate proportion visualization	Service or department proportions can be compared
TC7	Generate relationship visualization	Association between selected variables is visible
TC8	Perform correlation analysis	Correlation values and relationships are correctly displayed
TC9	Apply PCA	Data is successfully reduced and visualized
TC10	Apply t-SNE or UMAP	Alternative lower-dimensional visualization is generated

Students must provide:

- Test outputs.
- Screenshots.

- Generated visualizations.
- Interpretation of observations.

7. Performance Analysis

Students should evaluate the implementation using different dataset sizes where possible.

Suggested dataset sizes:

- 1,000 records.
- 5,000 records.
- 10,000 records.

METRIC	1,000 RECORDS	5,000 RECORDS	10,000 RECORDS
Dataset Loading Time			
Amount Visualization Generation Time			
Distribution Visualization Generation Time			
Correlation Analysis Time			
PCA Execution Time			
t-SNE / UMAP Execution Time			

Students must analyze:

- Effect of increasing dataset size.
 - Change in execution time.
 - Computational requirements.
 - Performance differences between PCA and the selected comparison technique.
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8. Analysis, Trade-offs and Justification

8.1 Visualization Selection

Students must analyze:

- Why was each visualization selected?
- Which data characteristics influenced the selection?
- Could another visualization represent the same information?

- Which visualization provides the clearest interpretation?
-

8.2 Distribution Visualization

Compare:

- Histogram versus density plot.
- Boxplot versus violin plot.

Students must discuss:

- Distribution information provided.
 - Spread and variation.
 - Group comparison.
 - Advantages and limitations.
-

8.3 Proportion Visualization

Compare suitable approaches such as:

- Pie chart versus side-by-side bar chart.
- Stacked bar chart versus treemap.

Students must justify which visualization provides the clearest representation for the selected healthcare data.

8.4 Relationship and Dimensionality Analysis

Students must compare:

- Scatterplot versus bubble chart.
- Scatterplot versus hexbin plot where applicable.
- Correlation matrix versus individual relationship plots.
- PCA versus t-SNE or UMAP.

Students should discuss:

- Patterns observed.
 - Differences between techniques.
 - Ease of interpretation.
 - Strengths and limitations.
-

8.5 Final Recommendation

Based on the results obtained, students must provide a final recommendation.

The recommendation should answer:

Which visualization and dimensionality reduction approaches are most suitable for analyzing the selected healthcare service and patient dataset, and why?

The answer must be justified using evidence from the implemented visualizations and analysis.

9. Broader Considerations / Professional Responsibility

Students must discuss the following considerations.

CONSIDERATION	DESCRIPTION
Data Integrity	Healthcare data must be represented accurately
Patient Privacy	Personally identifiable information must not be unnecessarily exposed
Responsible Interpretation	Conclusions must be supported by observed results
Documentation	Dataset, variables, techniques, and results must be clearly documented
Professional Responsibility	Data must be handled carefully because healthcare information can be sensitive

Students must demonstrate responsible data handling throughout the assignment.

10. Conclusion

The conclusion should summarize:

- The healthcare dataset analyzed.
- The visualization techniques applied.
- Important service and patient patterns identified.
- Application of CO1 concepts.
- Application of CO2 visualization techniques.

- Application of CO3 relationship and dimensionality reduction techniques.
- Comparison of alternative approaches.
- Important analytical decisions.
- Limitations.
- Possible future improvements.

The conclusion should explain how raw healthcare data was transformed into meaningful visual information.

11. Student Reflection

Students must answer the following questions:

1. What did you learn from this assignment beyond what was directly taught in the classroom?

Discuss your experience of handling and visualizing healthcare-related data.

2. Which visualization technique was the most challenging and why?

Explain the difficulties encountered.

3. What problem did you encounter while selecting a visualization, and how did you solve it?

Provide a specific example.

4. How did your understanding of data types influence your visualization decisions?

Explain how numerical and categorical data affected the selected charts.

5. Which visualization provided the most useful insight from the dataset?

Explain why.

6. How did PCA and t-SNE or UMAP change your understanding of multivariate healthcare data?

Discuss the patterns observed.

7. If you were given additional time and resources, what would you improve and why?

12. References

Students must provide references for:

1. Dataset source.
2. Python documentation.
3. Data handling libraries.
4. Visualization libraries.
5. Dimensionality reduction tools or documentation.
6. Academic or technical resources used.
7. Other sources referred to during implementation.

All references must be appropriately cited in the final report.

13. Assessment Rubric – 100 Marks

ASSESSMENT CRITERION	MARKS	KEY EVALUATION FOCUS
Problem Understanding & Formulation	10	Understanding of healthcare data challenges, dataset characteristics, requirements, and constraints
Application of Course Knowledge	20	Application of CO1, CO2, and CO3 concepts
Solution Methodology / Design / Implementation	20	Visualization workflow, design choices, methodology, and implementation
Modern Tools / Techniques	10	Python tools, visualization libraries, dimensionality reduction tools, and implementation evidence
Results, Testing & Validation	15	Test cases, generated outputs, validation, and interpretation

ASSESSMENT CRITERION	MARKS	KEY EVALUATION FOCUS
Analysis, Trade-offs & Justification	15	Comparison of techniques and evidence-based analytical decisions
Broader Considerations / Professional Responsibility	5	Data integrity, privacy, and responsible handling
Technical Documentation & Reflection	5	Report quality, references, screenshots, and reflection
TOTAL	100	

14. CO – PO – Assessment Mapping

ASSESSMENT COMPONENT	CO	BLOOM'S LEVEL	MARKS
Problem Formulation	CO1, CO2, CO3	L4 – Analyze	10
Application of Course Knowledge	CO1, CO2, CO3	L4 – Analyze	20
Solution / Design / Implementation	CO1, CO2, CO3	L5/L6 – Evaluate/Create	20
Modern Tool Usage	CO1, CO2, CO3	L3/L4 – Apply/Analyze	10
Validation	CO1, CO2, CO3	L4/L5 – Analyze/Evaluate	15
Analysis & Justification	CO1, CO2, CO3	L5 – Evaluate	15
Broader Considerations	CO1, CO2, CO3	L5 – Evaluate	5
Documentation & Reflection	CO1, CO2, CO3	L4 – Analyze	5
TOTAL	CO1, CO2, CO3		100

COMPLETED SOLUTIONS, PSEUDOCODE, AND REQUIRED GRAPHS

This section is appended to the original assignment brief. The original first-page and final-page tables have not been redesigned or re-aligned.

A. Dataset and Problem Answer

Question: What dataset was selected and why?

The UCI Diabetes 130-US Hospitals dataset was selected because it is a public, documented, multivariate and anonymized inpatient dataset containing categorical and integer variables. It has 101,766 encounters from 130 hospitals over 1999–2008 and 47 original features [1]. It supports the required amount, distribution, proportion, relationship, correlation, PCA and t-SNE analyses.

Question: What are the main limitations?

The dataset does not contain direct waiting-time, treatment-cost or patient-satisfaction fields. Therefore, the analysis uses available operational variables such as length of stay, laboratory procedures, medication count, previous visits and readmission outcome. Age is stored in ten-year intervals, so its midpoint is used only as an approximate numerical field. The results describe associations, not causal or clinical conclusions.

Pseudocode – complete analysis workflow

```
BEGIN
  LOAD diabetic_data.csv
  REPLACE placeholder "?" with missing values
  IDENTIFY categorical, ordinal, discrete and numeric variables
  CONVERT selected numeric columns to numeric type
  MAP age bands to ordered categories and approximate midpoints
  HANDLE missing values; exclude incomplete rows only for PCA/t-SNE
  CREATE amount charts: bar, grouped bar and heatmap
  CREATE distribution charts: histogram, boxplot and violin plot
  CREATE proportion charts: pie chart and bar chart
  CREATE relationship scatterplot and calculate Pearson correlations
  STANDARDIZE selected numeric variables
  FIT PCA with two components and record explained variance
  FIT t-SNE with fixed seed and documented perplexity
  PLOT PCA and t-SNE projections by readmission outcome
  INTERPRET patterns conservatively and document limitations/privacy controls
END
```

B. Data Handling and Visualization Fundamentals (CO1)

Variable type	Variables used	Visual mapping
Categorical	gender, race, specialty, readmitted, admission/discharge IDs	Labels and color; bar charts and legends.

Ordinal categorical	age bands	Ordered x-axis; age midpoint only for approximate numeric analysis.
Discrete integer	length of stay, lab procedures, procedures, medications, visit counts, diagnoses	Position on axes; bars, histograms, boxplots and scatterplots.
Identifiers	encounter_id, patient_nbr	Used for counts only; never displayed at record level.

Question: How were aesthetics, scales and color selected?

Position and aligned axes are used for primary comparisons because they are easier to judge accurately than area or angle. Color distinguishes readmission outcomes or categories. A sequential blue palette represents volume in the heatmap, while a diverging palette centered at zero represents correlation. Horizontal bars improve readability for long specialty labels.

C. Amount Visualization (CO2)

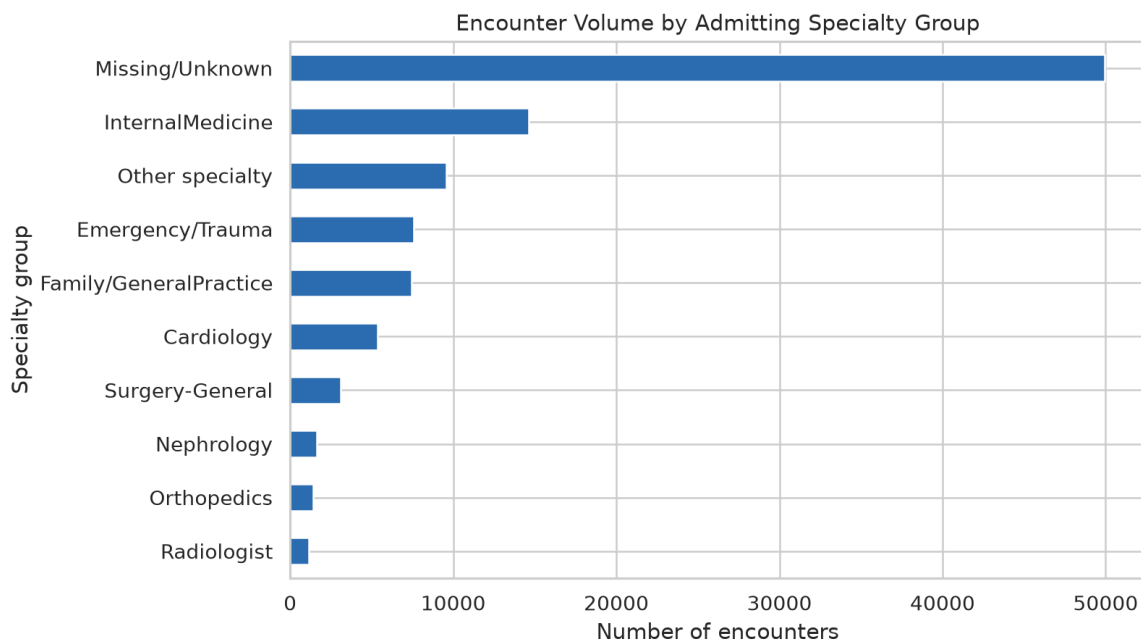


Figure A1. Encounter volume by admitting specialty group.

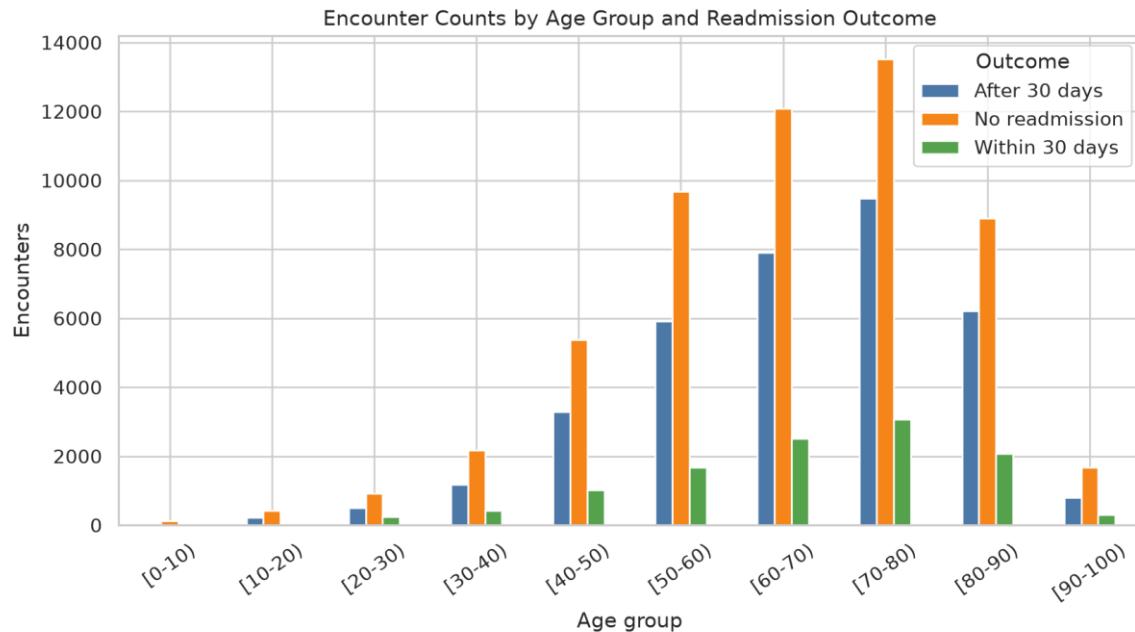


Figure A2. Encounter counts by age band and readmission outcome.

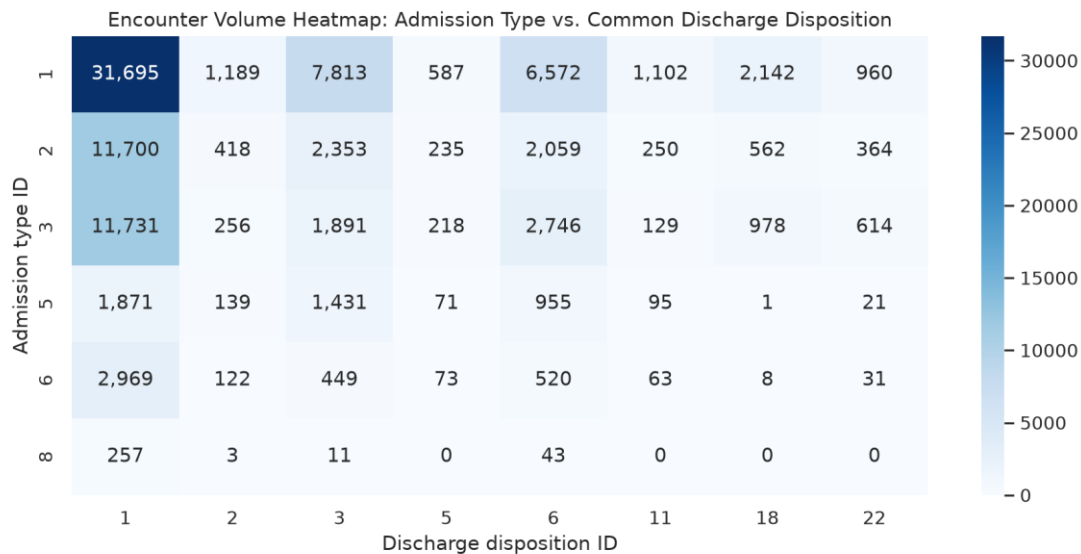


Figure A3. Encounter volume heatmap for admission and common discharge categories.

Question: What amount patterns were observed?

Encounters are concentrated in Internal Medicine and the consolidated Other specialty group. The 60–80 year age bands account for a large share of encounters. The admission/discharge heatmap reveals high-volume category intersections that could be investigated operationally.

Question: Which amount chart is clearest?

The horizontal bar chart is clearest for ranking specialty volumes because categories share one aligned quantitative scale. The grouped bar chart is better for comparing outcomes within age groups, while the heatmap is best for finding dense combinations but is less precise for exact comparison.

D. Distribution Analysis (CO2)

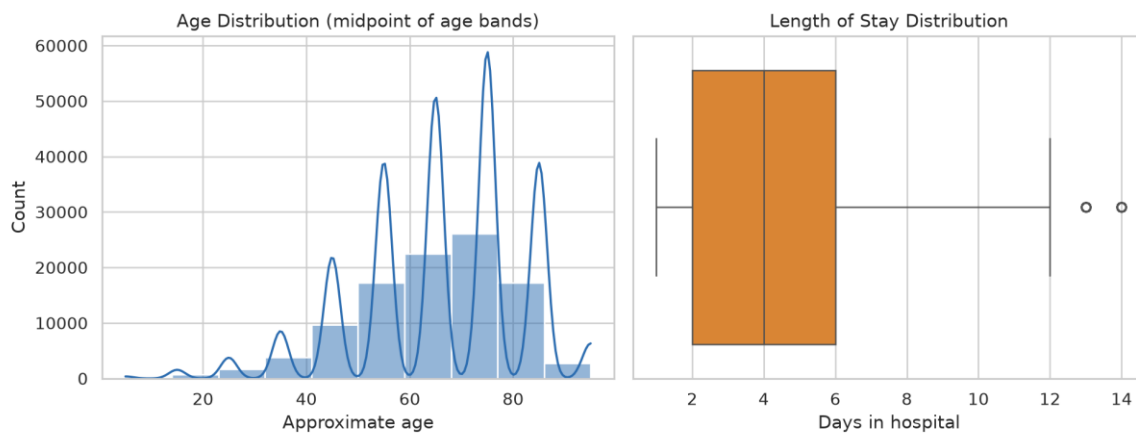


Figure A4. Approximate age histogram and length-of-stay boxplot.

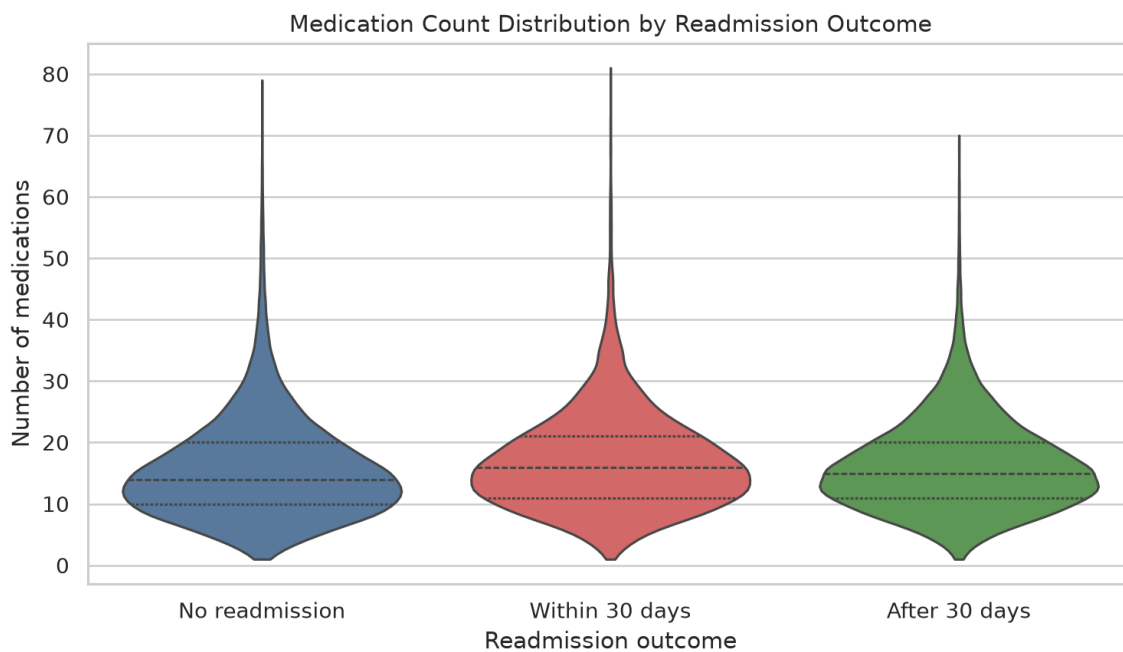


Figure A5. Medication-count distribution by readmission outcome.

Question: What distribution patterns, spread and outliers were identified?

The encounter population is concentrated in older adult age bands, especially around 60–80 years. Length of stay is discrete and right-skewed: most stays are short, with a small number extending to the upper permitted range. Medication counts have broad spread. Boxplots are strongest for medians and outliers; violin plots additionally show distribution shape.

E. Proportion Visualization (CO2)

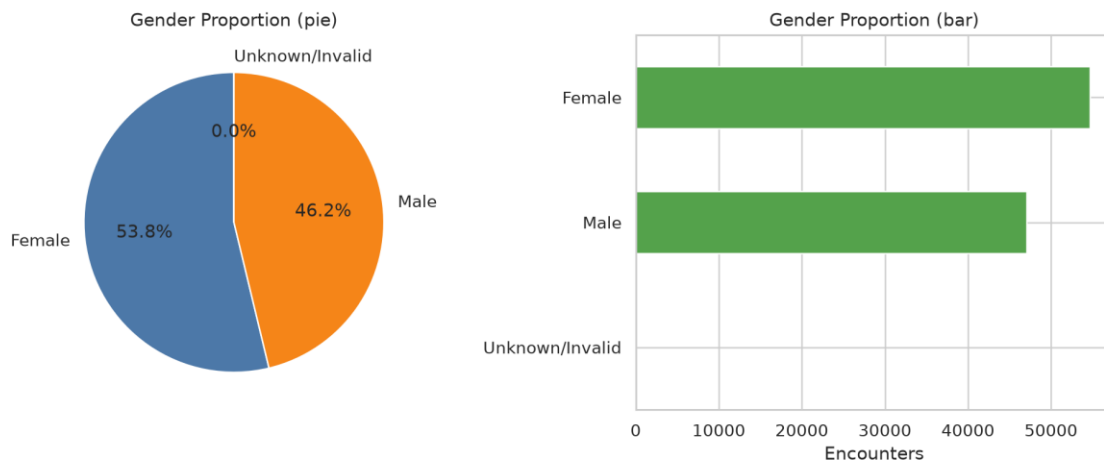


Figure A6. Gender proportions shown as a pie chart and a bar chart.

Question: Which proportion chart is clearer?

The bar chart is the final choice because aligned lengths support more accurate comparison. The pie chart provides a quick part-to-whole view, but it is less precise when comparing category sizes.

F. Relationship and Correlation Analysis (CO3)

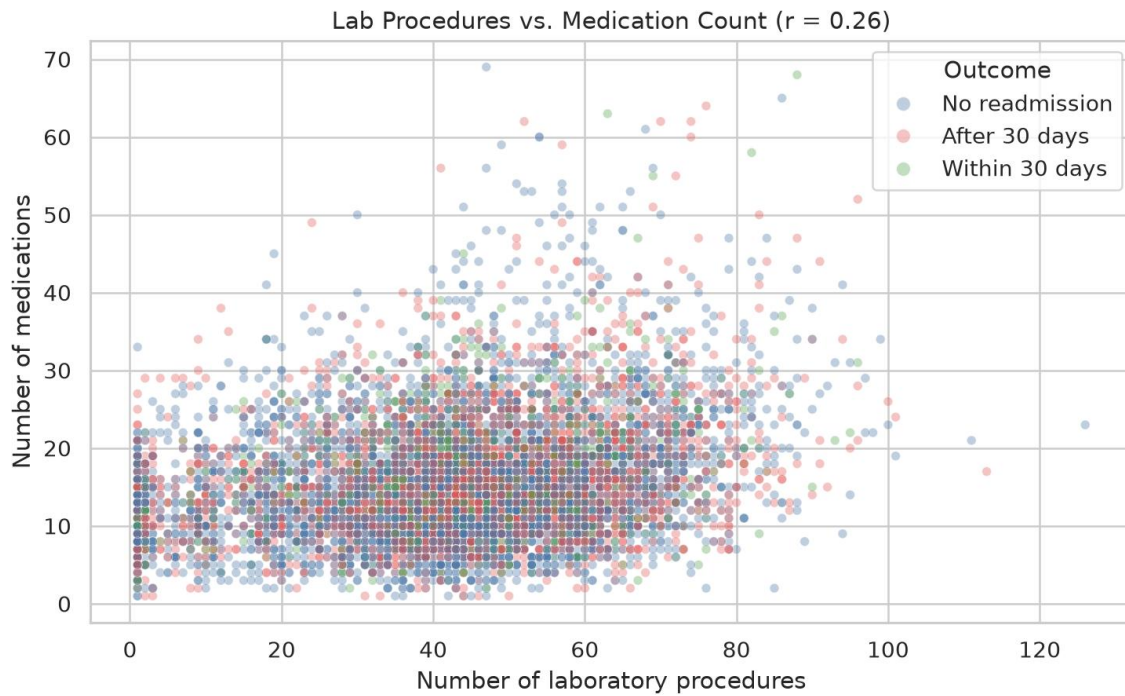


Figure A7. Laboratory procedures versus medication count, colored by readmission outcome.

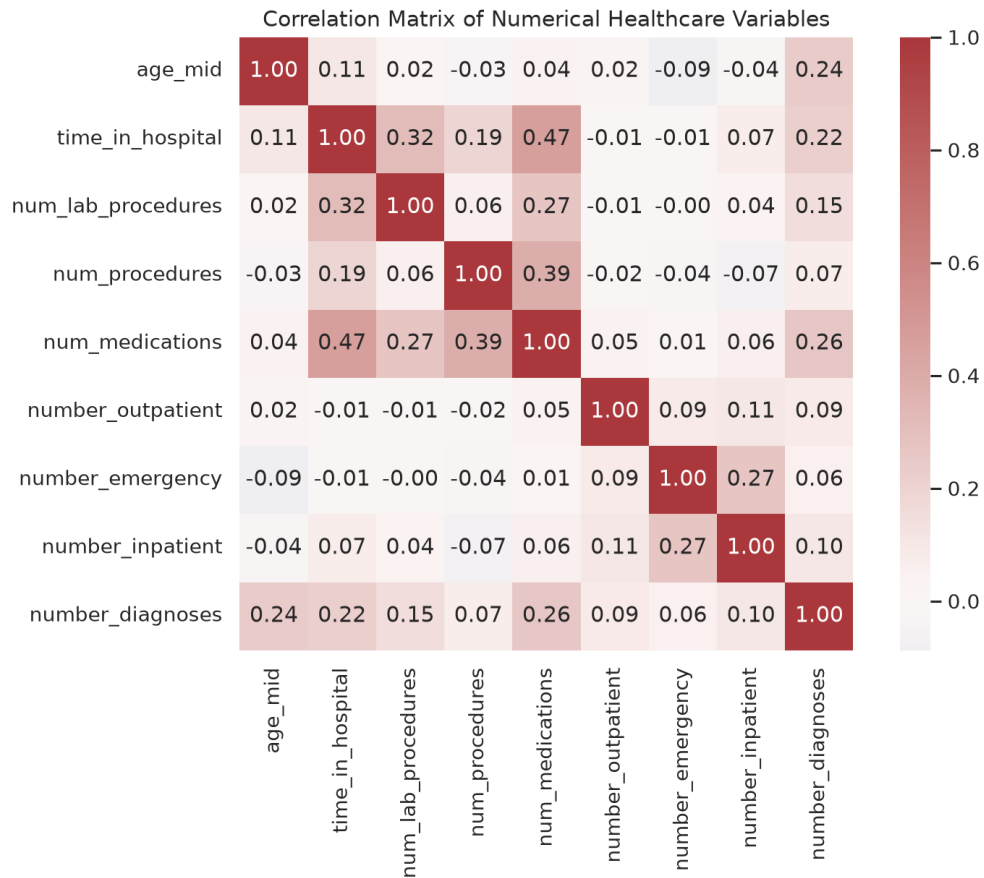


Figure A8. Pearson correlation matrix for selected numerical variables.

Question: What relationship was found?

Laboratory procedures and medication count have a positive but modest association, $r = 0.27$. The wide scatter and overlapping outcome colors show that laboratory activity alone does not determine medication count or readmission outcome.

Question: Which correlations are most important?

Length of stay and medication count show the strongest selected association, $r = 0.47$. Length of stay and laboratory procedures have $r = 0.32$, while laboratory procedures and medications have $r = 0.27$. These are positive associations, not evidence of causation.

Question: Why use a correlation heatmap?

It shows many pairwise coefficients in a compact view, uses color to reveal positive and negative association, and prints exact values so interpretation does not depend only on color intensity.

G. PCA and t-SNE (CO3)

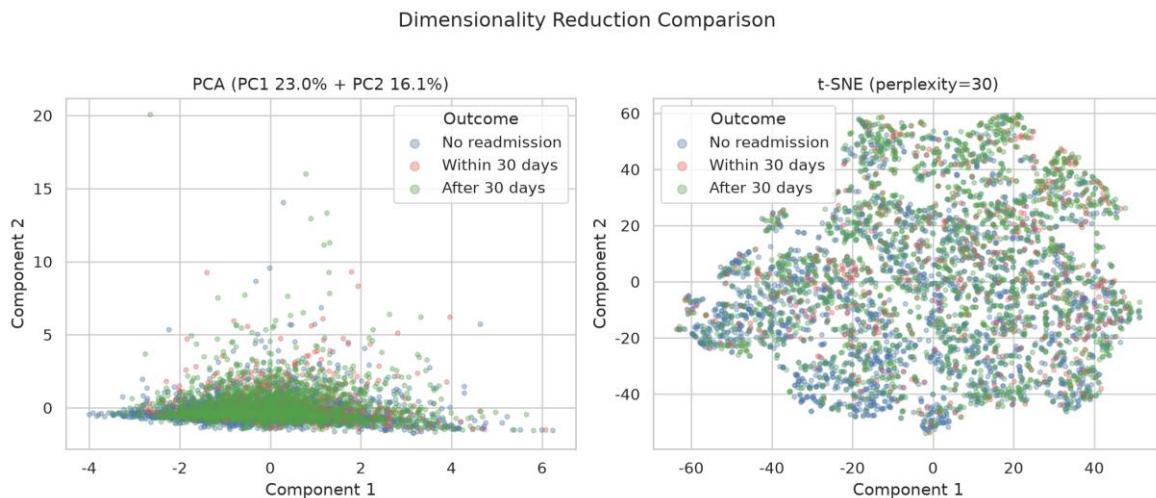


Figure A9. PCA and t-SNE projections of standardized multivariate healthcare variables.

Question: What did PCA show?

Nine standardized numerical variables were reduced to two components. The first two components retain 39.1% of standardized variance. Readmission groups overlap substantially, so the selected variables do not form clearly separated clusters in a two-dimensional linear projection.

Question: How does t-SNE compare with PCA?

PCA is interpretable, reproducible and reports explained variance. t-SNE can expose local nonlinear neighborhoods, but its axes have no direct meaning and its appearance depends on parameters and random seed. PCA is recommended for the main report, with t-SNE as a complementary exploratory view.

H. Results and Validation

Metric	Result
Rows / original columns	101,766 / 50 columns in the CSV including identifiers and target
Unique anonymized patients	71,518
Mean / median length of stay	4.40 / 4 days
Mean laboratory procedures	43.10
Median medications	15
No readmission / within 30 days / after 30 days	53.9% / 11.2% / 34.9%

Question: How was the solution validated?

The row count and source characteristics were checked against the UCI documentation. Numeric conversion, age ordering, missing-value handling, figure titles, axis labels and legends were checked. Fixed random seeds were used for sampled plots and t-SNE. Identifiers were excluded from visualization to protect privacy.

I. Final Recommendation, Responsibility, Conclusion and Reflection

Question: What is the final recommendation?

Use horizontal bars for rankings, grouped bars for subgroup comparisons, heatmaps for category intersections and correlations, histograms/boxplots for distributions, bar charts for proportions, scatterplots for individual relationships, PCA for the primary dimensionality-reduction view, and t-SNE only as an exploratory supplement. A future dashboard should show sample size and missingness for every filter.

Question: How was professional responsibility addressed?

The source file was preserved, missing markers were handled explicitly, patient and encounter identifiers were not displayed, and conclusions were limited to observed associations. The data contains sensitive demographic and clinical information, so any operational deployment would require appropriate institutional privacy and governance controls.

Question: What is the conclusion?

The analysis transforms anonymized inpatient records into interpretable views of volume, distributions, proportions, relationships and multivariate structure. The most notable numerical result is the moderate positive association between length of stay and medication count. The main limitation is the absence of direct waiting-time, cost and satisfaction measures.

Question: What was learned and what would be improved?

The work demonstrates that healthcare visualization requires careful treatment of missing-value codes, ordinal age bands, repeated patient encounters and privacy. t-SNE was the most challenging technique because its parameters affect visual neighborhoods. With more resources, the system should add cost, waiting-time and satisfaction variables, compare UMAP, provide uncertainty measures, test subgroup fairness and be evaluated by healthcare stakeholders.

J. References

[1] Clore, J., Cios, K., DeShazo, J., & Strack, B. (2014). Diabetes 130-US Hospitals for Years 1999–2008. UCI Machine Learning Repository. <https://doi.org/10.24432/C5230J>

[2] Strack, B. et al. (2014). Impact of HbA1c Measurement on Hospital Readmission Rates. BioMed Research International. <https://doi.org/10.1155/2014/781670>

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